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THE XP CURVES ORIGIN STORY

How the Progression System Was Built, Broken, and Rebuilt From Scratch

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CHAPTER 1: THE FIRST ATTEMPT — VIBES-BASED MATH

The XP system for Get Clocked started the way most game systems start: with gut feelings and round numbers. The very first version was born in an early MVP review session where the goal was simple — *"make the numbers feel good."*

The 100 XP Idea

The initial concept was straightforward. A typical 30-minute boxing session was sketched out with rough XP-per-minute rates: jump rope at 3 XP/min, shadowboxing at 4 XP/min, bag combos at 5 XP/min, pushups at 3, treadmill at 3, and rest at 0. Add it all up and a 30-minute workout produced roughly 102.5 XP. So the first proposal was: every level costs 100 XP. One session, one level. Clean and simple.

That lasted about 30 seconds before the response came back: "I meant 1,000 XP." Bigger numbers feel better. A jab should earn you points that feel meaningful, not pocket change. So the scale shifted — everything multiplied by 10, and 1,000 XP became the baseline.

The Pokémon Stagger

The next idea was to make levels progressively harder within each ranking tier, inspired by Pokémon's experience curves. Ten ranking tiers were created — Prospect, Scrapper, Fighter, Warrior, Gladiator, Veteran, Phenom, Icon, Apex, and GOAT — each spanning 10 levels. Within each ranking, levels would stagger up: Prospect Level 1→2 cost 500 XP, while Level 9→10 cost 2,300 XP. Then Scrapper would shift the entire curve upward, and so on.

The session targets were specified as a smooth ramp from 5 sessions (Prospect) to 30 sessions (GOAT), totaling 167 sessions for an entire Rookie prestige — about 10 months at 4x/week training.

The Prestige Layer

On top of the ranking stagger, a prestige multiplier was layered: Rookie at 1.0x, Beginner at 1.15x, Intermediate at 1.3x, Advanced at 1.5x, and Pro at 1.75x. This meant the final boss — Pro, GOAT, Level 99→100 — would cost roughly 14,175 XP, taking 14+ sessions (2-3 weeks) for a single level. Endgame grind.

The problem was that none of this was grounded in anything real. The "1,154 XP per session" number that the curves were calibrated to came from a **59-minute workout structure** that hadn't been built yet. The XP rates were guesses. The combo formula existed on paper but had never been calculated against actual punch combinations. Everything was vibes-based math — directionally right, but structurally hollow.

CHAPTER 2: THE ARCHITECTURE CHANGED UNDERNEATH

While the XP system sat on paper, the workout architecture was completely rebuilt. The original 59-minute, 15-cycle, 3-workouts-per-tier structure was scrapped in favor of a streamlined 35-minute (later 40-minute) single-loadout design with 10 boxing cycles, championship rounds, and equipment-dependent variations.

The New Workout Structure

Each tier now had one loadout workout built from:

Warmup: 3 minutes of jump rope (or high knees if no rope)

Grind: 10 boxing cycles, each containing 30s shadowboxing + 30s bag work (or 60s shadow if no bag) + 30s pushups + 30s rest

Championship Rounds: 2 rounds of freestyle shadowboxing at 2x XP

Conditioning: 120s curlups + 60s rest

Cooldown: 10 minutes on treadmill (or jogging in place)

The Combo System Took Shape

Each tier got specific combo pools with progressive difficulty. Rookie used simple 2-3 punch combos (1-2, 2-3, 1-2-5). Beginner introduced slips. Intermediate added rolls and pullback. Advanced wove in 5-9 move combinations with full defensive integration. Pro unleashed the complete arsenal — 7-11 move combinations using all punches (1-8, including the rarely-used lead and rear body hooks) plus every defensive move.

The combo XP formula was defined: Base Move Sum × Length Bonus × Defense Bonus × (Duration ÷ 30). Punch values ranged from 5 (jab) to 15 (rear body hook). Defense moves carried their own values. Length bonuses rewarded longer combinations. Defense bonuses rewarded combinations that integrated defensive movements.

The 666 XP Problem

When the actual XP output of the new 35-minute workout was finally calculated segment by segment — jump rope at 0.20/sec, shadowboxing at 0.33/sec, pushups at 0.22/sec, curlups at 0.18/sec, treadmill at 0.25/sec, rest at 0.01/sec — the full-equipment session produced roughly **666 XP**. The shadow-only session (no bag, no treadmill) produced about 465 XP.

The level curves had been calibrated to **1,154 XP per session** from the old 59-minute workout. The new workout produced 43% less XP. Every progression target was now wrong. A Rookie who was supposed to hit Level 100 in 35 sessions would now take roughly 60. The entire system was broken.

Three options were considered: inflate the XP rates (dishonest — a jab is a jab), lower the level thresholds (keeps rates honest), or accept slower progression. The decision was to recalibrate the curves to the real numbers, not the other way around.

CHAPTER 3: THE FIRST RECALIBRATION — TWO CONSTRAINTS

The first real math happened when the system was rebuilt using the formula XP(n) = BASE + COEFF × n^1.15, where n is the level number. The exponent 1.15 was chosen to create a gentle but consistent upward curve — each level costs slightly more than the last, but not so steeply that mid-game feels punishing.

The Two-Constraint Solver

Two constraints were set for each tier:

Constraint 1: Total sessions to Level 100. Rookie should take ~35 sessions, Beginner ~70, Intermediate ~100, Advanced ~138, Pro ~225. Total journey: ~575 sessions.

Constraint 2: First session level gain. A brand-new Rookie should hit Level 10 in their first session. A new Beginner should hit Level 6. Intermediate Level 4. Advanced Level 4. Pro Level 2.

With two constraints and two unknowns (BASE and COEFF), the system was solvable. A numerical solver was built in Python using scipy's fsolve function to find the exact BASE and COEFF values that satisfied both constraints simultaneously for each tier.

But there was still a foundational problem: the XP values being fed into the solver were the rough 666 XP estimates, not tier-specific calculations. A Rookie doing simple 1-2 combinations earns far less combo XP per round than a Pro doing 7-11 move defensive sequences. The session XP shouldn't be the same across tiers.

CHAPTER 4: STARTING FROM SCRATCH — THE LOADOUT REBUILD

This was the turning point. Instead of estimating, **every single combo in every tier's loadout was calculated individually**, using the exact XP formula, the exact punch values, and the exact workout structure.

The Granular XP Calculator

A Python script was built that walked through every segment of every tier's 40-minute loadout workout:

Warmup: Jump rope at 0.20 XP/sec for 180 seconds = 36.0 XP (identical across all tiers)

10 Rounds: Each round's combo was calculated using the actual combo pool. For shadow-only users, 60 seconds of combo XP. For bag users, 30 seconds shadow + 30 seconds bag. Pushups at 0.22/sec for 30 seconds = 6.6 XP. Rest at 0.01/sec for 30 seconds = 0.3 XP.

Championship Rounds: Freestyle shadowboxing at the tier's base rate, with 2x XP multiplier.

Conditioning: Curlups at 0.18/sec for 120 seconds = 21.6 XP. Rest at 0.01/sec for 60 seconds = 0.6 XP.

Cooldown: Treadmill at 0.25/sec for 600 seconds = 150.0 XP (or jog in place at 0.18/sec = 108.0 XP).

The Combo Breakdown

The key differentiator between tiers was the combo XP. Each tier's combo pools were loaded, and the average combo XP per 30-second segment was calculated by running every combo through the formula:

Combo XP = (Sum of Punch Values) × (Length Bonus) × (Defense Bonus) × (Duration ÷ 30)

Punch values: Jab (1) = 5, Cross (2) = 8, Lead Hook (3) = 10, Rear Hook (4) = 10, Lead Uppercut (5) = 12, Rear Uppercut (6) = 12, Lead Body Hook (7) = 13, Rear Body Hook (8) = 15. Defense moves like slip left, slip right, roll, and pullback added their own values on top.

The results were dramatic. The combo complexity difference between tiers created massive XP variance:

Tier	Avg Combo XP/Round	Shadow XP/Session	Description
Rookie	~17.8	679.8	Simple 2-3 punch combos (1-2, 2-3, 1-2-5)
Beginner	~25.3	829.8	Added slips (1-2-slip left, slip left-2-3)

Intermediate	~38.6	1,096.6	Rolls, pullback, longer sequences
Advanced	~69.1	1,706.3	5-9 move combos, full defense weaving
Pro	~93.5	2,291.0	7-11 move combos, complete arsenal

These numbers — **679.8, 829.8, 1,096.6, 1,706.3, and 2,291.0** — became the foundation everything else was built on. Not estimates. Not averages. Exact calculations from the actual workout content.

The Bag Differential

Bag users earned additional XP from two sources: jump rope replaced jog-in-place for the warmup (same 36 XP either way, but the XP rate was actually 0.20/sec for both), and the real difference came from treadmill vs. jog-in-place for cooldown: 150 XP vs. 108 XP. The net bag advantage was 45.6 XP/session, modest enough that shadow-only users weren't punished but meaningful enough that equipment investment felt worthwhile.

CHAPTER 5: THE CURVES THAT ACTUALLY WORKED

With real per-session XP values locked, the level curves could finally be solved properly. The session targets were refined to create a natural progression where each tier took meaningfully longer than the last:

Tier	Sessions to L100	Time @ 4x/week	1st Session Level
Rookie	21	1.2 months	L10
Beginner	49	2.8 months	L7
Intermediate	88	5.1 months	L6
Advanced	140	8.1 months	L5
Pro	212	1.0 year	L4

The Formula

The level cost formula remained XP(n) = BASE + COEFF × n^1.15, with Level 0→1 always costing just 1 XP (the instant hook — you level up almost immediately upon starting your first workout, creating an immediate dopamine hit).

The solver found these coefficients:

Tier	BASE	COEFF	EXP	Total XP Budget
Rookie	70.2	0.80	1.15	14,276
Beginner	124.6	3.08	1.15	40,660
Intermediate	188.5	8.48	1.15	96,501
Advanced	361.5	22.12	1.15	238,882
Pro	660.2	45.78	1.15	485,692

What the Numbers Mean

For Rookie (BASE=70.2, COEFF=0.80): Level 1→2 costs 71 XP. Level 50→51 costs 127 XP. Level 99→100 costs 222 XP. With 679.8 XP per session, a Rookie gains roughly 9-10 levels in their first session and gradually slows to about 3-4 levels per session by the end. The curve is gentle — a 3.2x ratio between L1 and L100 cost — because Rookie is the introductory tier. Players blow through it in 5 weeks.

For Pro (BASE=660.2, COEFF=45.78): Level 1→2 costs 706 XP. Level 50→51 costs 3,900 XP. Level 99→100 costs 7,620 XP. The curve is much steeper but the BASE dominates — even L1 costs a significant chunk of a session. Pro is a flat grind where you gain roughly 3-4 levels in session 1, then settle into +1 level per session for most of the journey. This was accepted as

fitting: by the time you reach Pro, you're a veteran. The game doesn't need to shower you with dopamine. You're just putting in work.

Bag Users

With bags adding 45.6 XP per session, the same curves produced slightly faster progression:

Tier	Shadow Sessions	Bag Sessions	Sessions Saved
Rookie	21	18	3
Beginner	49	43	6
Intermediate	88	80	8
Advanced	140	131	9
Pro	212	202	10

CHAPTER 6: THE L500 DETOUR — WHERE IT ALL WENT SIDEWAYS

With L100 curves locked, the next idea seemed logical: extend levels to 500, inspired by Call of Duty's prestige system. The concept was that at Level 100, players could choose to prestige (resetting to Level 0 at the next tier) or stay and grind toward Level 500 for exclusive badges and rewards. It was supposed to be the best of both worlds — Pokémon's progressive leveling meets CoD's prestige decision.

The Math That Couldn't Work

The goal was for L101-500 to take roughly 2x the time of L1-100. So if Pro takes 212 sessions to L100, it should take about 424 sessions total to L500. That means 400 additional levels in 212 extra sessions.

But the L99→L100 cost for Pro was already **9,690 XP**. With 2,291 XP per session, that's 4.2 sessions for a single level. If the curve continued upward — even by +1 XP per level — 400 more levels at ~9,690+ each would require roughly **1,700 additional sessions**, not 212.

The fundamental problem was explained through a staircase metaphor: by step 99, each step is 9,690 inches tall and you can only jump 2,291 inches per day. Adding 400 more steps that are the same height or taller means years of grinding. The only way to hit the 2x target was to make the steps shorter after L100 — an 87% cost reduction from L99's price.

The Solutions That All Failed

Hard reset: Drop cost back to L1 levels after prestige. Felt jarring and artificial — you just achieved something massive and suddenly levels cost nothing.

Soft reset: Drop to roughly L50 cost and climb slower. Still a visible cliff that breaks the feeling of one continuous journey.

Flat Phase 2: Every post-100 level costs the same fixed amount. For Pro, that meant each level at 1,169 XP — an 88% drop from L99. The math worked but the experience was described as "[redacted] " — going from the hardest level you've ever faced to something trivially easy.

L200 cap: Reducing the ceiling to 200 gave a cleaner 3x multiplier (Pro L200 = 637 sessions, about 3.1 years). But even this felt like an awkward compromise.

Piecewise curves: Hand-tuning separate formulas for L1-100 and L101-500 with different BASE/COEFF/EXP values. Created visible seams and felt engineered rather than natural.

Exponent changes: Switching from n^1.15 to n^1.3 to give Pro more curve shape. The result was almost identical — the BASE still dominated at 2,318 out of 2,336 total cost per level. No exponent could overcome that floor.

The Realization

After days of trying to make 500 levels work — pure power curves, three-constraint solvers, piecewise functions, stagnation targeting — the conclusion was blunt: *"once I introduced the 500 levels shit, it got* [redacted] *."* The L100 curves that had been locked before the detour were great. The problem wasn't the math — it was the requirement.

The decision was made: Level 100 is the cap. Post-100 progression lives in a badge system, not additional levels. Players who choose to stay at a tier instead of prestiging can farm exclusive badges and achievements. The levels stop, but the journey doesn't.

CHAPTER 7: THE V1 vs V2 SHOWDOWN

With L100 re-confirmed as the cap, one final question remained: how aggressively should the first session hook players? Two versions were compared side-by-side using an interactive React visualization.

V1: Original (L10 → L4)

Session 1 levels: Rookie hits L10, Beginner L7, Intermediate L6, Advanced L5, Pro L4. Wide range, maximum dopamine for new Rookies. The tradeoff was a very flat Rookie curve — L1 costs 143, L100 costs 358, only a 2.5x ratio. But Rookie is only 21 sessions long. Nobody notices the flatness when they blow through the entire tier in 5 weeks.

V2: Compressed (L5 → L3)

Session 1 levels: Rookie hits L5, down to Pro at L3. Tighter band, more uniform feel. But the math was devastating — forcing Rookie into L5 with only 30k total XP meant every level cost roughly 300 XP with nowhere to go. L1 costs 208, L100 costs 328, a pathetic 1.6x ratio. It wasn't a curve; it was a flat line. No escalation, no feeling of increasing challenge. Just counting.

The Verdict

V1 won, and it wasn't close. Seven level-ups popping in 40 minutes is a dopamine firehose — every few minutes you hear a ding, see a new number, feel like you're flying. That's what gets someone to come back for session 2. The flatness of Rookie's curve is invisible because the tier is over in weeks. The tiers where steepness matters — Advanced (8 months) and Pro (1 year) — both versions had strong escalation.

V1's final locked power curve form used the formula cost(n) = A × n^B, with Level 0→1 always at 1 XP:

Tier	A	B	L1 Cost	L100 Cost	L100/L1 Ratio	Total
Rookie	142.70	0.200	143	358	2.5×	30,000
Beginner	149.31	0.4261	149	534	3.6×	75,000
Intermediate	243.88	0.5139	244	1,031	4.2×	173,000
Advanced	443.04	0.5185	443	1,899	4.3×	320,000
Pro	651.85	0.5337	652	2,950	4.5×	500,000

CHAPTER 8: THE COMPLETE PICTURE

The Full XP Stack

The locked curves represent the baseline — what you earn from running your loadout workout with zero bonuses. On top of this foundation, the full XP stack layers additional systems:

Streak Multipliers: 0-2 days = 1.0x, ramping to 3.0x at 365+ consecutive days. This is the primary long-term accelerator.

Time Multipliers: In-workout multipliers that increase the longer your session runs, rewarding extended training.

Session Completion Bonus: Finishing a full workout with no skips = +10% XP.

Championship Rounds: Rounds 11+ earn 2x XP, rewarding players who push past the standard 10-round structure.

Double XP Events: Monthly weekend events, random daily windows, and Combo of the Day challenges.

Comeback Mechanics: Players who break a streak get a Redemption Mode with temporarily boosted XP to prevent permanent disengagement.

The BMF Endgame

Beyond the 5 prestige tiers lies the ultimate achievement: BMF (Baddest Motherfucker) status, requiring 250 Pro-level sessions completed. Beyond that is BMFK (Baddest Motherfucking Killer), demanding 365 consecutive days of Pro-level training. These are not levels — they're lifetime achievements that only the most dedicated fighters will ever unlock. At the curve's pace, BMF requires roughly 2+ years of consistent Pro training. BMFK adds a full year of unbroken daily commitment on top of that.

What Made It Work

The final system works because it was built from the ground up on real data, not estimates. Every number traces back to a specific combo pool, a specific punch value, a specific segment duration. The progression curves weren't designed in a vacuum and then forced onto the workout system — the workout system was built first, measured precisely, and the curves were derived from those measurements.

The journey from vibes-based 100 XP levels to a mathematically grounded power curve system involved:

4 major rewrites of the XP architecture

3 different level cap experiments (100, 200, 500)

6+ failed curve fitting approaches for extended levels

1 complete from-scratch rebuild grounded in actual workout data

2 final versions compared through interactive visualization

The result is a system where a Rookie's first session is a dopamine firehose (L0 to L10 in 40 minutes), the middle tiers provide a satisfying but lengthening grind, and Pro demands a full year of disciplined training to complete — exactly the kind of earned achievement that makes a fighter feel like they've actually accomplished something.

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